

Exercise 1: Implementing the Singleton Pattern

Logger.java:

```
public class Logger
{
    private static Logger obj;
    private Logger(){
        System.out.println("Logger Instance Created");
    }
    public static Logger getInstance(){
        if(obj == null){
            obj = new Logger();
        }
        return obj;
    }
    public void log(String message){
        System.out.println("Log: " + message);
    }
}
```

Main.java:

```
public class Main
{
    public static void main(String[] args)
    {
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();
        logger1.log("Application Started");
        logger2.log("User Logged In");
        if(logger1==logger2){
            System.out.println("Both references point to the same object");
        }
        else{
            System.out.println("Different objects created");
        }
    }
}
```

Output:

```
First Message
Second Message
Only One Object Created

...Program finished with exit code 0
Press ENTER to exit console. □
```